Supplementary Materials: Analysis of Arousal Ratings

Missing Values

Of the 614,400 LLM arousal ratings (64 conditions × 96 expressions × 100 iterations), 25 were missing with gpt-oss (20b) due to internal errors returned by Ollama during prompt processing. All other invalid values resulted from violations of the requested response format: gemma3 (4b): 1; gemma3 (12b, 27b): 0; llama3.1 (8b): 61; phi3 (3.8b): 79; phi3 (14b): 30; phi4 (14b): 8; gpt-oss (20b): 146. These values were more frequent with the 1-9 scale ($N = 198$) than with the 0–100 scale ($N = 127$), thereby not replicating the valence results in this regard. By example type, there were 67 (none), 65 (low only), 100 (high only), and 93 (both) invalid values. Overall, missing values made up only 0.06% of all arousal ratings.

Variance Decomposition by Method Factors

To address aim (1), we first rescaled ratings from the 1–9 format to the 0–100 scale to avoid artificial differences between response formats and then fitted the mixed model specified in the main text. Table S1 shows the estimated variance components and their percentages. Since we aimed to investigate consistency—whether the ordering of stimuli was stable across models, response formats, and prompt examples—we were predominantly interested in the interactions involving the stimulus factor, that is, the relative error variance components.

The stimulus factor accounted for the largest proportion of variance (52.53%) in LLM arousal ratings. This is desirable, as actual differences in the arousal of multi-word expressions perceived by humans (i.e., the universe score variance to be measured) should have the largest effect on LLM ratings. Still, Table S1 shows that a considerable portion of the observed variance (almost 25%) was attributable to methodological facets (method variance). The residual variance component (22.68%) was larger than for valence ratings, indicating an increased degree of variability of ratings across iterations within each cell of the design.

In the D study, we estimated the coefficients $E\rho^2$ and Φ for our design with 8 model conditions, 4 example conditions, and 2 response format conditions. The

generalizability coefficient $E\rho^2 = .96$ corresponds to the expected correlation between arousal ratings of two independent studies with the same stimuli but using different methodological facets (Brennan, 2001), indicating excellent reliability. The dependability coefficient was only slightly lower ($\Phi = .93$), indicating that absolute arousal ratings for the stimuli (averaged across facets) were reliable. The D study above assumes a single iteration per cell. Beyond the analyses we preregistered, we considered the number of iterations in an additional analysis for another D study, leading to a shrinkage of the residual term, resulting in $E\rho^2 = .97$ and $\Phi = .94$. Given the already high coefficients, accounting for the 100 iterations led to only marginal improvements here, although averaging across iterations generally reduces error variance in LLM simulations (see also the aggregation analysis below). Finally, we conducted another D study where the number of facets per factor was set to $n = 1$, yielding $E\rho^2 = .58$, indicating the expected correlation between two different LLM-rating conditions across the investigated stimuli

(moreover, $\Phi = .53$).

Table S1
Variance components in LLM arousal ratings based on a mixed model.

Effect	$\hat{\sigma}^2$	Percent
Stimulus	316.55	52.53
Model	9.79	1.63
Format	2.18	0.36
Example	29.74	4.94
Stimulus × Model	52.45	8.70
Stimulus × Format	0.69	0.12
Stimulus × Example	6.44	1.07
Model × Format	7.41	1.23
Model × Example	7.59	1.26
Format × Example	0.00	0.00
Stimulus × Model × Format	9.34	1.55
Stimulus × Model × Example	11.35	1.88
Stimulus × Format × Example	0.06	0.01
Model × Format × Example	1.78	0.30
Stimulus × Model × Format × Example	10.55	1.75
Residual	136.63	22.68

Note. The universe score variance and the relative error variance components (part of $\sigma^2(\delta)$) are highlighted in bold. The variance component for Format × Example was estimated at the boundary of zero, so lme4 reported a singular fit. Because the model converged, we retained the full model rather than simplifying it as described in the preregistration for convergence problems.

Correlation between Human and LLM Ratings

To address aim (2), we computed the correlation between human and LLM arousal ratings across the 96 stimuli for each of the 64 conditions (Figure S1; the dashed line indicates the correlation of $r = .92$ obtained with GPT-4o by Martínez et al., 2024).

As correlations are bounded by the reliability of the variables, we also assessed the reliability of the human benchmark ratings of arousal with an analogous generalizability analysis. In the G study, a crossed design with random intercepts for stimuli, raters, and their interaction (the residual) attributed 31.24%, 27.54%, and 41.22% of the variance in human ratings to these components, respectively. Note that the rater variance component is larger for arousal ratings than for ratings of valence reported in the main manuscript. In the D study, the generalizability coefficient was computed as,

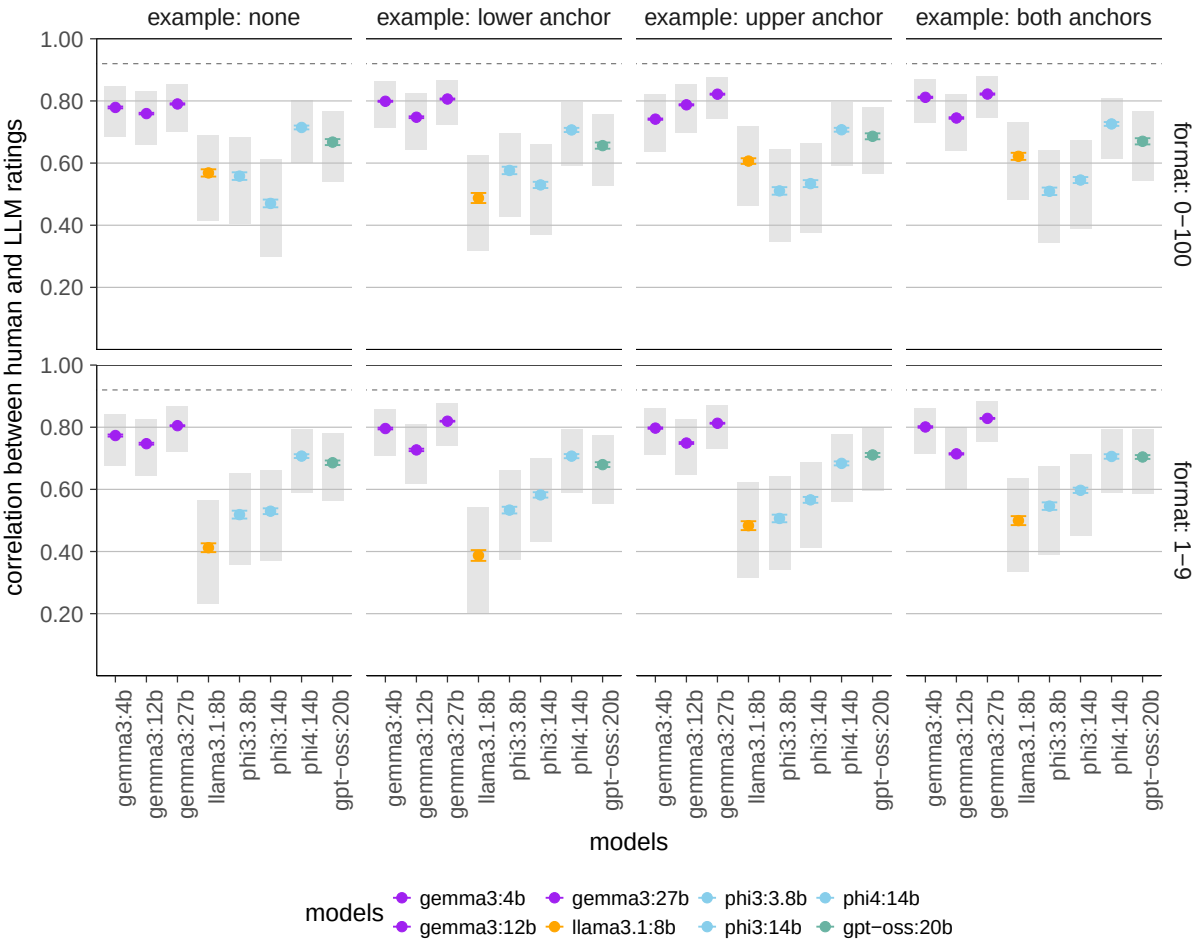
$$E\rho^2 = \frac{\sigma^2(\tau)}{\sigma^2(\tau) + \frac{\sigma^2(sr)}{n'_r}}, \tag{1}$$

where $\sigma^2(sr)$ is the stimulus-by-rater (residual) component and n'_r the number of raters. The resulting $E\rho^2 = .95$ indicates excellent reliability of the mean arousal ratings across the 23 raters in Martínez et al. (2024): The mean item ratings from two independent sets of 23 raters would be expected to correlate at $r = .95$. y Error bars and shaded areas are defined as in the main text. Figure S1 shows that many of the gemma3 configurations could achieve an adequate $\widehat{IRR}$. However, all correlations with human ratings were lower than those shown in Martínez et al. (2024). Many of the models showed a correlation with human ratings of arousal that was defined to be below the acceptable threshold ($\widehat{IRR} < .70$). Figure S2 shows the LLM simulation condition with the highest correlation between LLM and human ratings of arousal. To this end, Figure S2 LLM ratings were averaged across iterations for each stimulus.

To assess whether the performance ranking of LLM simulation conditions generalizes to new stimuli, we conducted a split-half analysis (exploratory). We randomly split the 96 stimuli into two halves of 48 stimuli each. Within each half, we computed the correlation between human and LLM ratings for each condition and iteration and

Figure S1

Correlation between human and LLM ratings of arousal for multi-word expressions



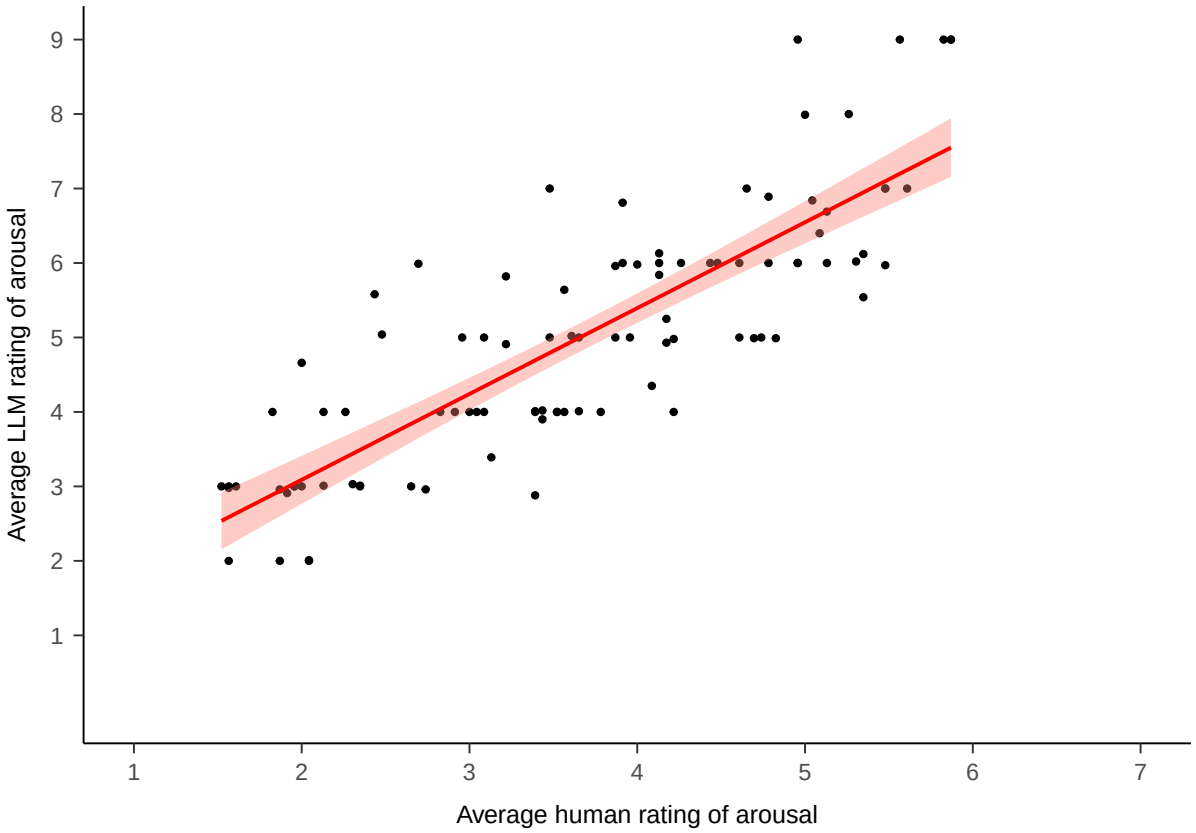
Note. The dashed horizontal line indicates the correlation between human and LLM ratings obtained with GPT-4o by Martínez et al. (2024). Colored error bars (dark gray in print) indicate 95% confidence bounds based on the MCSE. The gray shaded areas indicate the 95% confidence interval for each correlation coefficient, averaged across simulation iterations in each condition.

averaged these correlations across iterations, yielding 64 average correlations per half (one per condition). We then computed the Pearson and Spearman correlations between the two halves across the 64 conditions. Repeating this procedure for 1,000 random splits yielded a mean split-half correlation of $r = .93$ [.86, .97] and a mean rank correlation of $r_s = .92$ [.85, .96] (values within square brackets indicate 95% percentiles). Applying the Spearman–Brown formula to estimate the stability for the full set of 96 stimuli yielded $r = .96$ and $r_s = .96$. Thus, the relative performance of the LLM simulation conditions was highly stable across independent sets of stimuli.

We also examined whether averaging repeated LLM ratings, which reduces their

Figure S2

Highest correlation between human and LLM ratings of arousal for multi-word expressions

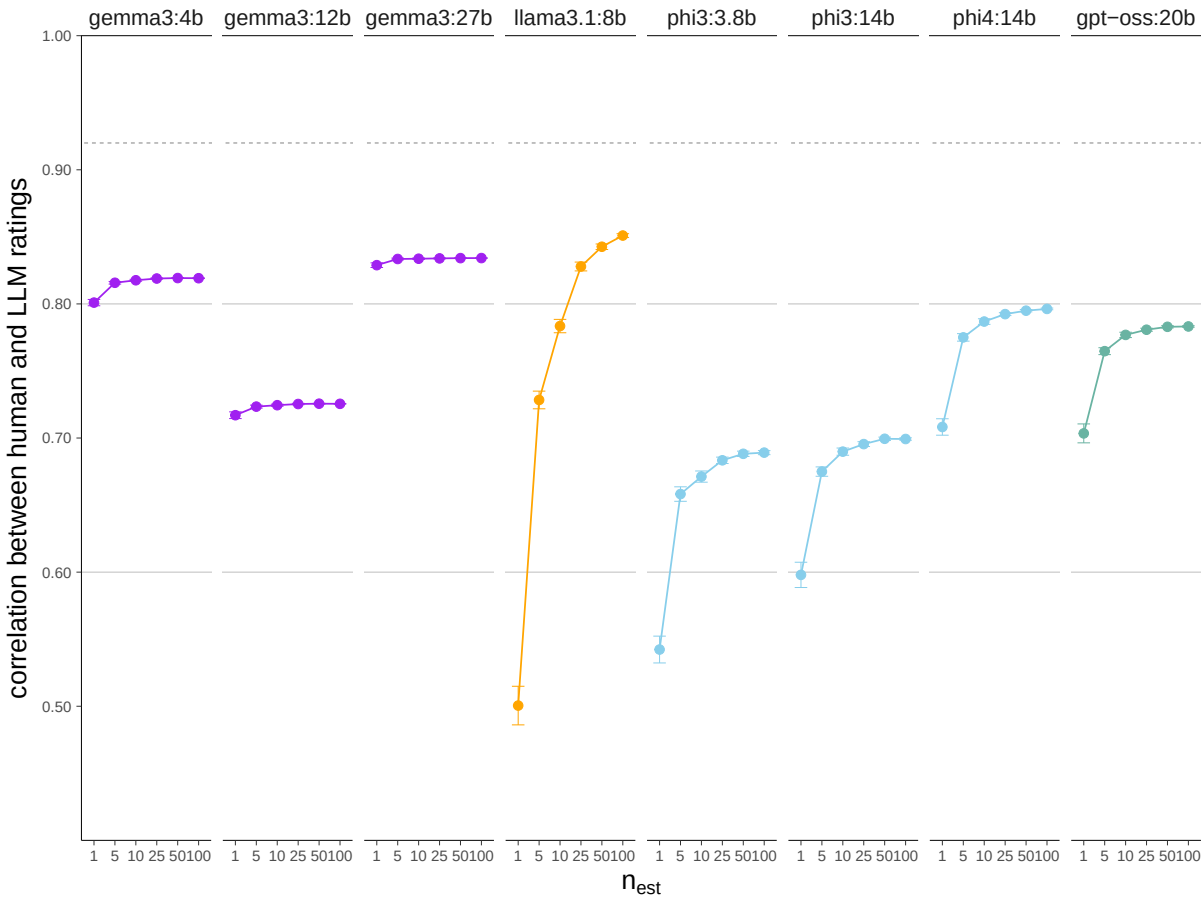


Note. The scatter plot shows the condition with the highest correlation between human and LLM ratings (gemma3 (27b) with a 1–9 response format and examples at both anchors, i.e., low and high arousal). To visualize results, stimulus ratings were aggregated across iterations. The shaded band indicates the 95% confidence interval of the regression line.

stochasticity, improves the correlation with human ratings. Using the bootstrap procedure described in the Performance Measures section (level b), we varied the number of averaged ratings $n_{\text{est}} \in \{1, 5, 10, 25, 50, 100\}$, where $n_{\text{est}} = 1$ corresponds to Figure S1. Figure S3 shows the results for the condition implementing the original prompt of Martínez et al. (2024) (1–9 format, examples of low and high arousal). Aggregation substantially improved performance for smaller models such as llama3.1 (8b), whose $\widehat{IRR}$ increased from approximately .50 to over .80 when averaging across 100 iterations, comparable to the larger models. In contrast, the gemma3 models (especially 27b), whose unaggregated ratings already correlated around .80 with human ratings, improved only slightly. In the example LLM simulation condition, llama3.1 even showed the highest correlation with human ratings compared to all other models.

Figure S3

Correlation between human and LLM ratings of arousal for different levels of aggregation



Note. The plot shows the condition with both types of examples and a 1-9 scale. The dashed, horizontal line indicates the correlation between human and LLM ratings obtained with GPT-4o by Martínez et al. (2024). Colored error bars (dark gray in print) indicate 95% confidence bounds based on the MCSE.

Supplementary Materials: Analysis of temperature parameter settings

To verify which temperature was in effect in the main simulation, we ran all eight prompt configurations for the valence ratings using the four models llama3.1:8b, phi3:3.8b, phi3:14b, and phi4:14b, as the Ollama documentation indicates that the default model temperature is 0.8 when no model-specific default exists. However, as we accessed the models in Ollama 0.12.9 through `ellmer` (Wickham et al., 2025), Ollama overrides this default with the value 1.0 if the OpenAI-compatible endpoint is used (see the appendix of the main paper). Therefore, all models should have been run with the temperature 1.0. To test this empirically, we ran all 8 prompt configurations (4 example types $\times$ 2 response scales) for each of the four models with three temperature parameter settings: 1) default temperature (no explicit specification of this parameter), 2) temperature = 0.8, and 3) temperature = 1.0. Each condition in the design grid was run with 20 iterations (R 4.6.0, ellmer 0.4.0). Subsequently, we computed the entropy across the 20 responses for each stimulus per condition defined as $H(X) = -\sum_{x_i} p(x_i) \cdot \log_2 p(x_i)$ where $p(x_i)$ are the relative frequencies of the occurred responses. The average entropy was then computed for each combination of temperature and model. As the temperature parameter in an LLM is a means of directly manipulating the entropy of the response vector, lower values of the temperature should lead to lower entropy, whereas higher values should lead to higher entropy in the response distribution. The findings of this analysis can be seen in Table S2, showing that the default temperature of the models was in fact 1.0 and not 0.8.

Table S2

Mean entropy of ratings per stimulus by model and temperature setting (averaged across the 8 prompt conditions)

Model	Temperature	Entropy
llama3.1:8b	0.8	2.58 [2.52, 2.63]
llama3.1:8b	1.0	2.79 [2.74, 2.85]
llama3.1:8b	default	2.79 [2.74, 2.85]
phi3:3.8b	0.8	1.42 [1.38, 1.47]
phi3:3.8b	1.0	1.70 [1.66, 1.75]
phi3:3.8b	default	1.70 [1.65, 1.75]
phi3:14b	0.8	1.51 [1.46, 1.56]
phi3:14b	1.0	1.81 [1.76, 1.86]
phi3:14b	default	1.80 [1.74, 1.85]
phi4:14b	0.8	1.44 [1.38, 1.49]
phi4:14b	1.0	1.67 [1.62, 1.73]
phi4:14b	default	1.67 [1.61, 1.73]

Note. The *default* condition indicates the LLM simulation procedure used in the main manuscript where no temperature was specified. Values within square brackets indicate 95% confidence intervals across stimuli and prompt conditions.

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